Mathematical Programming:
Modelling and Some Issues

Sensitivity: Internal

Contents

- Closed Feasible Region (less than or equal to)
- Summation via Matrix Operations
- Chocolate Box Problem
- Interpretation/Opinion/Preference
- Extending the Model
- Additional Modelling Tips

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Less than or equal to

Note that when we use inequality constraints we must use $\leq$ or $\geq$

If you use a $<$ then the feasible set is not closed.

Consider:

maximise x
subject to:
$x < 1$

What is the maximum?

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Constraints via Matrix Operations

$$\sum_{i=1}^{n} a_i x_i$$

Let $a = (a_1, a_2, \ldots, a_n)^{\mathrm{T}}$ and $x = (x_1, x_2, \ldots, x_n)^T$

$$a^T x = (a_1, \ldots, a_n) \begin{pmatrix} x_1 \\ \vdots \\ x_n \end{pmatrix}$$

$$= \sum_{i=1}^{n} a_i x_i$$

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Linear Expressions in Pyomo

```
# Define integer variables
model.x = Var(range(4), domain=NonNegativeReals)

print(10*model.x[0] + 6*model.x[1] + 8*model.x[2] + 16*model.x[3])

cost = [10, 6, 8, 16]
print(sum(cost[i] * model.x[i] for i in model.x))
```

Define either manually or using `sum`

Both print out the following:

```
10*x[0] + 6*x[1] + 8*x[2] + 16*x[3]

10*x[0] + 6*x[1] + 8*x[2] + 16*x[3]
```

e.g.

$$10x_1 + 6x_2 + 8x_3 + 16x_4$$

Let $a = (10 \quad 6 \quad 8 \quad 16)^{\mathrm{T}}$ and $x = (x_1 \quad x_2 \quad x_3 \quad x_4)^{\mathrm{T}}$

$$a^T x = (10 \quad 6 \quad 8 \quad 16)\begin{pmatrix} x_1 \\ x_2 \\ x_3 \\ x_4 \end{pmatrix}$$

$$= 10x_1 + 6x_2 + 8x_3 + 16x_4$$

$$= \sum_{i=1}^{n} a_i x_i$$

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Linear Constraints in Pyomo

If we have the following constraint.

$$10x_1 + 6x_2 + 8x_3 + 16x_4 \geq 120$$

We can implement that in Python directly

```
# Define integer variables
model.x = Var(range(4), domain=NonNegativeReals)
# define a vector
cost = [10, 6, 8, 16]
# define a linear inequality constraint
model.constraint1 = Constraint(expr=sum(cost[i] * model.x[i] for i in model.x) >= 120)
# pretty print the constraint
model.constraint1.pprint()
```

Output:

```
constraint1 : Size=1, Index=None, Active=True
    Key  : Lower : Body                                  : Upper : Active
    None : 120.0 : 10*x[0] + 6*x[1] + 8*x[2] + 16*x[3] :  +Inf :   True
```

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Diet Problem (Week 2 Tutorial Q6a)

minimise $10x_1 + 6x_2 + 8x_3 + 16x_4 + 8x_5 + 3x_6 + 6x_7 + 8x_8 + 12x_9 + 12x_{10}$

subject to:

$18x_1 + 8x_2 + 22x_3 + 6x_4 + 8x_7 + 4x_8 + 12x_{10} \geq 100$ (Vitamin A Constraint)

$4x_1 + 6x_2 + 10x_3 + 12x_4 + 14x_5 + 6x_9 + 20x_{10} \geq 120$ (Vitamin B Constraint)

$22x_2 + 20x_6 + 8x_8 + 10x_9 \geq 200$ (Vitamin C Constraint)

$6x_1 + 14x_2 + 6x_4 + 10x_5 + 8x_6 + 6x_7 + 12x_9 + 4x_{10} \geq 80$ (Vitamin D Constraint)

$8x_3 + 6x_6 + 20x_7 + 40x_8 \leq 300$ (Fat Constraint)

and

$x_1, x_2, \dots, x_{10} \geq 0$ (Non-negativity)

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Diet Problem (Week 2 Tutorial Q6a)

Food	Vitamin A	Vitamin B	Vitamin C	Vitamin D	Fat	Cost	Attractiveness
1	18	4	-	6	-	10	6
2	8	6	22	14	-	6	1
3	22	10	-	-	8	8	5
4	6	12	-	6	-	16	5
5	-	14	-	10	-	8	6
6	-	-	20	8	6	3	4
7	8	-	-	6	20	6	10
8	4	-	8	-	40	8	10
9	-	6	10	12	-	12	6
10	12	20	-	4	-	12	7

	Minimum	Maximum
Vitamin A	100	-
Vitamin B	120	-
Vitamin C	200	-
Vitamin D	80	-
Fat	-	300

x_i - Number of food i
v_i^A - Amount of vitamin A in food i
v_i^B - Amount of vitamin B in food i
v_i^C - Amount of vitamin C in food i
v_i^D - Amount of vitamin D in food i
f_i - Amount of fat in food i
c_i - Cost of food i
a_i - Attractiveness of food i

minimise

$$\sum_i c_i x_i$$

subject to

$$\sum_i v_i^A x_i \geq 100$$

$$\sum_i v_i^B x_i \geq 120$$

$$\sum_i v_i^C x_i \geq 200$$

$$\sum_i v_i^D x_i \geq 80$$

$$\sum_i f_i x_i \leq 300$$

$$x_i \geq 0, \qquad \forall i$$

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Q6a Solution in Pyomo

minimise

$$\sum_i c_i x_i$$

subject to

$$\sum_i v_i^A x_i \geq 100$$

$$\sum_i v_i^B x_i \geq 120$$

$$\sum_i v_i^C x_i \geq 200$$

$$\sum_i v_i^D x_i \geq 80$$

$$\sum_i f_i x_i \leq 300$$

$$x_i \geq 0, \qquad \forall i$$

```python
from pyomo.environ import *

# Define the cost, vitamin, and fat values as lists
c = [10, 6, 8, 16, 8, 3, 6, 8, 12, 12]
vita = [18, 8, 22, 6, 0, 0, 8, 4, 0, 12]
vitb = [4, 6, 10, 12, 14, 0, 0, 0, 6, 20]
vitc = [0, 22, 0, 0, 0, 20, 0, 8, 10, 0]
vitd = [6, 14, 0, 6, 10, 8, 6, 0, 12, 4]
fat = [0, 0, 8, 0, 0, 6, 20, 40, 0, 0]

no_vars = len(c)

# Create a concrete model
model = ConcreteModel()

# Define the decision variables
model.x = Var(range(no_vars), domain=NonNegativeReals)

# Define the objective function
model.obj = Objective(expr=sum(c[i] * model.x[i] for i in range(no_vars)), sense=minimize)

# Define constraints
model.vita_min = Constraint(expr=sum(vita[i] * model.x[i] for i in range(no_vars)) >= 100)
model.vitb_min = Constraint(expr=sum(vitb[i] * model.x[i] for i in range(no_vars)) >= 120)
model.vitc_min = Constraint(expr=sum(vitc[i] * model.x[i] for i in range(no_vars)) >= 200)
model.vitd_min = Constraint(expr=sum(vitd[i] * model.x[i] for i in range(no_vars)) >= 80)
model.fat_max = Constraint(expr=sum(fat[i] * model.x[i] for i in range(no_vars)) <= 300)

# Solve the optimization problem
solver = SolverFactory('glpk')
solver.solve(model, tee=False)  # Set tee=True to display solver output

model.display()
```

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Chocolate Box Tutorial Question

Candy - x_i	Fat (g) - F_i	Calories – K_i	Weight (g) - W_i	Cost - C_i	Popularity - P_i
Buttery butterscotch	20	100	24	12	10
Caramel softie	16	100	22	12	10
Orange dream	10	80	15	12	8
Coconut Ice	7	50	12	15	9
Lemon sorbet	6	60	10	13	5
Lime sherbet	5	40	10	13	6
Strawberry surprise	13	80	14	16	7
Raspberry mousse	15	90	17	16	8
Ginger toffee	18	80	20	15	10
Rose-water posy	-	145	8	10	3
Walnut whirl	19	120	20	18	8
Grape bunch	-	120	15	20	4

- The selection chosen must weigh at least 320g
- Total fat content by weight must be at most 45% of the total weight of the selection.
- Calorific value of the selection must be at most 2850 calories.

<u>Minimise the cost of the selection</u>

minimise

$$\sum_i C_i x_i$$

subject to:

$$\sum_i W_i x_i \geq 320$$

$$\sum_i K_i x_i \leq 2850$$

$$\sum_i F_i x_i \leq 0.45 \sum W_i x_i$$

$$x_i \geq 0, \ \forall i$$
$$x_i \in \mathbb{Z}, \ \forall i$$

<u>Maximise the popularity of the selection</u>

Replace the objective function

maximise

$$\sum_i P_i x_i$$

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Chocolate Box Tutorial Results

Candy - x_i	Minimise Cost	Maximise Popularity
Buttery butterscotch	0	0
Caramel softie	9	0
Orange dream	0	0
Coconut Ice	0	5
Lemon sorbet	0	0
Lime sherbet	0	50
Strawberry surprise	0	0
Raspberry mousse	0	0
Ginger toffee	0	0
Rose-water posy	10	0
Walnut whirl	0	0
Grape bunch	3	5
Objective Value	268	365

What do we think to these selections?

Issues

- Variety
- Weight of maximising popularity is 635g. Nearly double the minimum weight of 320g
- What if we care about both cost and popularity?

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Interpretation

Your interpretation of the problem is important!

If you miss something in the model it can lead to answers that don't make sense in the real world.

Your interpretation of the solution is important!

Often the answers point out the issues in your model.

Thus, modelling is an iterative process, you should keep improving your model to better reflect reality where possible.

Better models lead to better predictions and decisions!

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Opinion and Preference

Your opinion and preferences matter!

Perhaps you/business really needs to save money, so you decide minimising cost is important.

Maybe you aren’t a fan of variety and would prefer a smaller selection, or the selection chooses your favourite chocolate. Be aware of personal bias!

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Sometimes there is more to the story

Let’s revisit the previous comment.

Perhaps you/business really needs to save money, so you decide minimising cost is important.

What if this leads to a substandard product and this fails to sell. In this case you haven’t necessarily saved money, but lost it!

Modelling well, whether it be Linear Programming, Simulation, Optimisation is hard!

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Extending the Model

Back to the chocolate box!

Issues

- Variety
- Weight of maximising popularity is 635g. Nearly double the minimum weight of 320g
- What if we care about both cost and popularity?

We will deal with variety in the next session, we can force the selection to pick every item a number of times and certain combinations of items.

An obvious extension to our model is to put a cap on weight. e.g. 1.5 times the minimum.

$$\sum_i W_i x_i \geq 320$$

$$\sum_i W_i x_i \leq 1.5 * 320$$

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Chocolate Box Tutorial Results

No Weight Cap

Candy - x_i	Minimise Cost	Maximise Popularity
Buttery butterscotch	0	0
Caramel softie	9	0
Orange dream	0	0
Coconut Ice	0	5
Lemon sorbet	0	0
Lime sherbet	0	50
Strawberry surprise	0	0
Raspberry mousse	0	0
Ginger toffee	0	0
Rose-water posy	10	0
Walnut whirl	0	0
Grape bunch	3	5

With Weight Cap

Candy - x_i	Minimise Cost	Maximise Popularity
Buttery butterscotch	0	0
Caramel softie	9	0
Orange dream	0	0
Coconut Ice	0	30
Lemon sorbet	0	0
Lime sherbet	0	0
Strawberry surprise	0	0
Raspberry mousse	0	0
Ginger toffee	0	0
Rose-water posy	10	5
Walnut whirl	0	0
Grape bunch	3	5

Why hasn't it affected the Minimise Cost Model?

Because the weight of the optimal solution was already less than the cap!

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Extending the Model

Issues

- Variety
- Weight of maximising popularity is 635g. Nearly double the minimum weight of 320g
- What if we care about both cost and popularity?

How to deal with both cost and popularity?

Ideas?

Combine the objectives

- Ratio
- Weighted Objective

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Ratio Objective

Candy - x_i	Cost - C_i	Popularity - P_i	$\frac{P_i}{C_i}$
Buttery butterscotch	12	10	0.833333
Caramel softie	12	10	0.833333
Orange dream	12	8	0.666667
Coconut Ice	15	9	0.6
Lemon sorbet	13	5	0.384615
Lime sherbet	13	6	0.461538
Strawberry surprise	16	7	0.4375
Raspberry mousse	16	8	0.5
Ginger toffee	15	10	0.666667
Rose-water posy	10	3	0.3
Walnut whirl	18	8	0.444444
Grape bunch	20	4	0.2

Maximise the sum of each candy's popularity per unit cost

maximise

$$\sum_i \frac{P_i}{C_i} x_i$$

subject to:

$$\sum_i W_i x_i \geq 320$$

$$\sum_i W_i x_i \leq 1.5 * 320$$

$$\sum_i K_i x_i \leq 2850$$

$$\sum_i F_i x_i \leq 0.45 \sum W_i x_i$$

$$x_i \geq 0, \ \forall i$$
$$x_i \in \mathbb{Z}, \ \forall i$$

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Result

Candy - x_i	Fat (g) - F_i	Calories – K_i	Weight (g) - W_i	Cost - C_i	Popularity - P_i	$\frac{P_i}{C_i}$	Maximise $\frac{P_i}{C_i}$
Buttery butterscotch	20	100	24	12	10	0.833333	0
Caramel softie	16	100	22	12	10	0.833333	0
Orange dream	10	80	15	12	8	0.666667	0
Coconut Ice	7	50	12	15	9	0.6	8
Lemon sorbet	6	60	10	13	5	0.384615	0
Lime sherbet	5	40	10	13	6	0.461538	32
Strawberry surprise	13	80	14	16	7	0.4375	0
Raspberry mousse	15	90	17	16	8	0.5	0
Ginger toffee	18	80	20	15	10	0.666667	0
Rose-water posy	-	145	8	10	3	0.3	8
Walnut whirl	19	120	20	18	8	0.444444	0
Grape bunch	-	120	15	20	4	0.2	0

Issue:

What if we care more about popularity than cost?

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Weighted Objective

Candy - x_i	Cost - C_i	Popularity - P_i
Buttery butterscotch	12	10
Caramel softie	12	10
Orange dream	12	8
Coconut Ice	15	9
Lemon sorbet	13	5
Lime sherbet	13	6
Strawberry surprise	16	7
Raspberry mousse	16	8
Ginger toffee	15	10
Rose-water posy	10	3
Walnut whirl	18	8
Grape bunch	20	4

Issue

- Cost and Popularity may be of a different scale
- What if cost was in thousands of pounds?

Combine objectives:

maximise $\sum_i P_i x_i$ and minimise $\sum_i C_i x_i$

We know that:

minimise $\sum_i C_i x_i \quad \Leftrightarrow \quad -(\text{maximise } -\sum_i C_i x_i)$

So:

maximise $\sum_i P_i x_i - \sum_i C_i x_i$

However, this gives an equal preference to both.

Weighted objective

maximise $(1-\alpha)\sum_i P_i x_i + \alpha(-\sum_i C_i x_i)$

maximise $(1-\alpha)\sum_i P_i x_i - \alpha\sum_i C_i x_i$

For a $40/60$ split (40% popularity, 60% cost) $\alpha = 0.6$

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Normalised Data

Candy - x_i	Cost - $\frac{C_i}{\max_i\{C_i\}}$	Popularity – $\frac{P_i}{\max_i\{P_i\}}$	$\frac{P_i}{\max_i\{P_i\}} - \frac{C_i}{\max_i\{C_i\}}$
Buttery butterscotch	0.6	1	0.4
Caramel softie	0.6	1	0.4
Orange dream	0.6	0.8	0.2
Coconut Ice	0.75	0.9	0.15
Lemon sorbet	0.65	0.5	-0.15
Lime sherbet	0.65	0.6	-0.05
Strawberry surprise	0.8	0.7	-0.1
Raspberry mousse	0.8	0.8	0
Ginger toffee	0.75	1	0.25
Rose-water posy	0.5	0.3	-0.2
Walnut whirl	0.9	0.8	-0.1
Grape bunch	1	0.4	-0.6

<u>Maximise the normalised weighted objective</u>

maximise

$$(1-\alpha)\sum_i \frac{P_i x_i}{\max_i\{P_i\}} - \alpha \sum_i \frac{C_i x_i}{\max_i\{C_i\}}$$

subject to:

$$\sum_i W_i x_i \geq 320$$

$$\sum_i W_i x_i \leq 1.5 * 320$$

$$\sum_i K_i x_i \leq 2850$$

$$\sum_i F_i x_i \leq 0.45 \sum W_i x_i$$

$$x_i \geq 0, \ \forall i$$
$$x_i \in \mathbb{Z}, \ \forall i$$

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Result - Equal Preference $\alpha = 0.5$

Candy - x_i	Fat (g) - F_i	Calories – K_i	Weight (g) - W_i	Cost - $\frac{C_i}{\max_i\{C_i\}}$	Popularity – $\frac{P_i}{\max_i\{P_i\}}$	$\frac{P_i}{\max_i\{P_i\}} - \frac{C_i}{\max_i\{C_i\}}$	Maximise Weighted Objective
Buttery butterscotch	20	100	24	0.6	1	0.4	0
Caramel softie	16	100	22	0.6	1	0.4	0
Orange dream	10	80	15	0.6	0.8	0.2	0
Coconut Ice	7	50	12	0.75	0.9	0.15	24
Lemon sorbet	6	60	10	0.65	0.5	-0.15	0
Lime sherbet	5	40	10	0.65	0.6	-0.05	0
Strawberry surprise	13	80	14	0.8	0.7	-0.1	0
Raspberry mousse	15	90	17	0.8	0.8	0	0
Ginger toffee	18	80	20	0.75	1	0.25	0
Rose-water posy	-	145	8	0.5	0.3	-0.2	11
Walnut whirl	19	120	20	0.9	0.8	-0.1	0
Grape bunch	-	120	15	1	0.4	-0.6	0

Selection has:

- a total weight of 376g
- 2795 kcals
- 168g fat

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Results for all capped models

Candy - x_i	Minimise Cost	Maximise Popularity	Maximise $\frac{P_i}{C_i}$	Maximise Weighted Objective (Equal)	Maximise Weighted Objective $\alpha = 0.59$	Maximise Weighted Objective $\alpha = 0.16$
Buttery butterscotch	0	0	0	0	0	0
Caramel softie	9	0	0	0	9	0
Orange dream	0	0	0	0	0	0
Coconut Ice	0	30	8	24	0	30
Lemon sorbet	0	0	0	0	0	0
Lime sherbet	0	0	32	0	0	0
Strawberry surprise	0	0	0	0	0	0
Raspberry mousse	0	0	0	0	0	0
Ginger toffee	0	0	0	0	0	0
Rose-water posy	10	5	8	11	10	5
Walnut whirl	0	0	0	0	0	0
Grape bunch	3	5	0	0	3	5

Completely different solutions for the first 3 columns!

Values of $\alpha \gtrsim 0.59$ return the Minimise cost model
Values of $\alpha < 1/6$ (≈ 0.167) return the Maximise popularity model

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Solve the Linear Program First

Solve the Linear Program first, Integer Program second.

An optimal solution for the LP model may exist on an integer solution.
i.e. you might just get lucky!

In that case you don’t need to solve the Integer Program (can be slow to solve for large models)

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Modelling a Linear Program

1. Write down what you know!
2. Identify the variables
3. Identify the objective
4. Identify the constraints
5. Are the constraints and objective linear?
6. Use appropriate notation.
   - e.g. label vitamin A content of item i - v_i^A
7. Are the variables Real or Integer?
8. Formulate the Model

2 and 3 are not always obvious. E.g. conflicting preferences, opinions, etc.

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Additional Tips

Model a Scaled/Simpler Problem

- It is easier to spot errors

Do you have test data?

- What if you already make a given profit and your model does worse?

Under Modelling

- Your model misses important features/constraints

Over Modelling

- Just like we extended, sometimes our models can be too restrictive
- We can also end up including redundant features/constraints
- We can also end up with conflicting constraints – Infeasible
  - Sales wants more than can be produced
- Sometimes an expert knows best

Please have a read of the R.E.D. Woolsey articles. They are both insightful and amusing.

- We will go over the Inflexible Steel Company in another session

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